Skin Lesion Classification: Milestone Report

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Abstract

I fine-tune an ImageNet-pretrained ResNet-50 to classify dermoscopic lesions as malignant or benign and compare it with dermatologists from Tschandl et al. [2019] on the same ISIC 2018 test images. Across three seeds, test AUROC is 0.929 ± 0.003, and at the readers' 78% specificity the model's sensitivity is $90.7\% \pm 1.4$ versus 76% for the mean reader. The threshold chosen on development data drifted on the test set (specificity 73.7%). Skin-tone and image-quality evaluation come next.

1 Domain question, intended use, and revisions

Dermatologists today examine skin lesions and decide whether to biopsy it or monitor it guided by criteria such as the ABCDE and the 7-point checklist. Identification mistakes delay treatment or cause unnecessary biopsy. The intended use is for clinicians to supports to their decision to biopsy or monitor. Previous research has shown that dermatology AI models degrade on darker skin tones [Daneshjou et al., 2022].

Current thesis. For dermatologists evaluating suspicious skin lesions, I test whether a fine-tuned image classifier detects malignant lesions, as well as human readers on expert-labeled dermoscopic images, and whether that performance holds for variations in skin tones and image quality.

Revisions since the proposal.

(1) The proposal planned to test on a portion of HAM10000, which would only have given a rough comparison with the reader study. Instead, I tested on the ISIC 2018 test set [Codella et al., 2019] which the reader classified. HAM10000 is now used only for training and validation.

(2) The proposal called the baseline "dermatologists", but the 76%/78% figures are the mean of 511 readers with varying experience, so instead they are called human readers.

(3) "Similar specificity" is now defined where the threshold is set on validation to give 78% specificity and applied unchanged to the test set. Then, sensitivity at matched test specificity is reported separately.

(4) Each model is trained with three seeds with results reported as mean $\pm$ SD.

2 What has been implemented

Data. HAM10000 [Tschandl et al., 2018] contains 10,015 dermoscopic images of seven diagnoses. From Tschandl et al. [2019], they are melanoma, basal cell carcinoma (BCC), and actinic keratosis/intraepithelial carcinoma being malignant. Nevi, benign keratosis, dermatofibroma, and vascular lesions are benign. I split HAM10000 into 8,584 training images (with 1,675 malignant) and 1,431 validation images (with 279 malignant). Lesions with multiple photos stayed in the same set. The ISIC 2018 test set has 1,512 images (with 307 malignant).

Preprint.

Model and training. I started with a ResNet-50 [He et al., 2016] pretrained on ImageNet, and replaced its last layer with one malignant score. Then, fine-tuned all layers for 10 epochs with AdamW with learning rate 10^{-4}, and batch size 32. Training images were randomly cropped, flipped, and rotated. Because only about 1 in 5 training images is malignant, missed malignant cases are weighted about 4 times more in the loss. We keep the epoch with the best validation AUROC. The cutoff for calling a lesion malignant is set on the validation set to give 78% specificity, matching the readers, and is then used unchanged on the test set. Training was run three times with different seeds.

3 Baseline and preliminary results

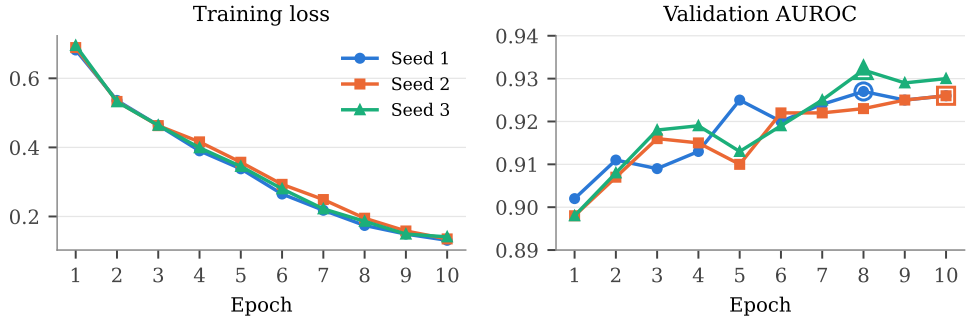
Baseline. In Tschandl et al. [2019], 511 human readers labeled the ISIC 2018 test images. Grouped into malignant versus benign, the average reader caught 76% of malignant lesions (sensitivity) while correctly catching 78% of benign ones (specificity). Combining all readers by majority vote gave an AUROC of 0.958.

Table 1: Malignant-versus-benign results, mean ± SD over three seeds. Validation: HAM10000 (1,431 images). Test: ISIC 2018 (1,512 images). "Fixed cutoff" uses the cutoff chosen on the validation set. "Matched specificity" picks the cutoff on the test set so specificity equals the readers' 78%.

	AUROC	Sensitivity	Specificity
Validation, fixed cutoff	0.928 ± 0.003	0.918 ± 0.015	0.781 ± 0.002
Test, fixed cutoff	0.929 ± 0.003	0.920 ± 0.018	0.737 ± 0.006
Test, matched specificity	0.929 ± 0.003	0.907 ± 0.014	0.78–0.80
Human readers [Tschandl et al., 2019]	0.958^*	0.76	0.78

*AUROC of all readers combined by majority vote.

Figure 1: Training loss and validation AUROC per epoch for three seeds. Doubled markers show the epoch. Validation AUROC plateaus near 0.92–0.93 from epoch 5 while training loss keeps falling.



Observations and failed attempts. (1) On a 500-image subset, validation AUROC peaked at 0.786 on epoch 5 and fell to 0.756 by epoch 10, but training loss kept falling. For this reason, the best epich was chosen by validation AUROC. With the full dataset, AUROC plateaued instead of declining. (Figure 1).

(2) The cutoffs that matched the readers' 78% specificity were 0.04–0.11, far from the usual 0.5, so the model's scores work for setting a cutoff but are not the chance of cancer.

(3) The cutoff did not carry over exactly. With the same cutoff, the model correctly called 78% of benign lesions benign on validation, but only 73.7% on test. The model separated malignant from benign lesions just as well on both sets (AUROC 0.929 vs. 0.928), but more benign test lesions scored above the cutoff, so more were incorrectly flagged as malignant.

(4) Melanoma had the lowest sensitivity of the three malignant types for every model on the validation test sets. On test, sensitivity was 0.873 ± 0.027 for melanoma, 0.978 ± 0.019 for BCC, and 0.977 ± 0.023 for actinic keratosis.

4 What the evidence supports and what remains unresolved

On clean dermoscopic images from the reader study, the model's sensitivity is about 15 points higher than the mean reader's at the same specificity. Even the weakest of the three trained models is about 14 points higher. Therefore, the proposal's revision rule (sensitivity more than 10 points below the readers) is not triggered.

This result has four limits. First, with the cutoff chosen in advance, part of the gain comes from extra false alarms. Compared with the mean reader, the model catches about 49 more of the 307 cancers but also flags about 52 more of the 1,205 benign lesions. Second, the mean reader averages readers of different experience, and when all readers are combined by majority vote, they still outperform the model (AUROC 0.958 vs. 0.929). Third, the $\pm$ values show only how much results change between training runs; they do not show how much results would change with a different set of test images. Confidence intervals for that will be added. Fourth, the per-type results cannot be compared with the readers' per-type results. The readers had to name the exact diagnosis, while the model only had to flag a lesion as malignant.

What is still unresolved is whether this performance holds on darker skin and lower-quality images and if the cutoff shifts more on those images like on the test set.

5 Next: distribution shift and stress conditions

Skin tone. Fitzpatrick17k [Groh et al., 2021] contains regular clinical photos (not dermatoscope close-ups), each labeled with a Fitzpatrick skin type estimated by annotators from the photo. Using only lesions labeled as benign or malignant, I will compare lighter skin (types I–II) with darker skin (types V–VI) within this dataset. Because all these photos are the same kind, any difference comes from skin tone rather than photo type. I will report the number of images in each group and confidence intervals,

Image quality. I will blur the ISIC 2018 test images and make them darker or brighter, each at three levels of strength. At each level, I will compare AUROC, sensitivity, and specificity (using the same fixed cutoff) with the results on the original images. The thesis will be narrowed if sensitivity drops by 10 or more points on darker skin (types V–VI) or at any level of image damage.

6 Class-survey case summary and comparison-matrix entry

Case summary. Dermatologists deciding whether to biopsy a suspicious skin lesion. A fine-tuned ResNet-50 is compared with 511 human readers on the ISIC 2018 test images used in their study, and then evaluated under skin-tone and image-quality shift. Main limitation: evidence comes from curated image datasets and cannot show performance in a clinic.

Domain	User / decision	AI system	Baseline	Evidence	Shift	Current finding
Dermatology	Dermatologist: biopsy or monitor	Fine-tuned ResNet-50	Human readers (76% sens. at 78% spec.)	HAM10000, ISIC 2018 test, Fitzpatrick17k	Skin tone, image quality	Sens. 90.7% at matched spec. on clean images; threshold drifted on test; shift results pending

7 Plan for completion

(1) Add bootstrap confidence intervals, and download Fitzpatrick17k and count its images by skin type and label.

(2) Run the image-quality tests and write up their results.

(3) Run the skin-tone evaluation and write up its results.

(4) Analyze missed melanomas and cutoff drift, and finish the final report.

8 Contributions

This project is being done individually.

9 Declaration of LLM usage

Generative AI was used to explain, debug, draft, and edit the project code and the milestone report.

Code

`https://github.com/jinn2024S/mlproj_milestone`

References

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